

Construction of the ACE Descriptor in TRACE-v2

A stage-by-stage walkthrough of `flashace/physics.py`

Implementation notes

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Abstract

This note documents how the atomic cluster expansion (ACE) descriptor is built in the TRACE-v2 code, mapping each block of `ACEV2Descriptor` in `flashace/physics.py` onto the corresponding equation of the manuscript and, where relevant, onto Varshalovich, Moskalev & Khersonskii, *Quantum Theory of Angular Momentum* (cited as VMK ch.sec(eq)). Concrete tensor shapes are given for the production configuration ($r_c = 6$ Å, $\ell_{\max} = 2$, $N_r = 12$, hidden = 64, correlation channels = 16, correlation order = 4).

1 Overview

The descriptor runs in six stages. Stages 1–3 are per *edge*; stage 4 reduces edges to atoms; stages 5–6 build the per-atom centre state that the attention block later refines.

Stage	Object	Manuscript	Output shape
0	irrep bookkeeping	—	—
1	radial basis $B_n(d)$	Eqs. (7)–(9)	$[L, 12]$
2	spherical harmonics $Y_{\ell m}$	Eq. (11)	$[L, 9]$
3	edge tensor $\mathbf{a}_{ij}$	Eqs. (11)–(12)	$[L, 60]$
4	neighbour density $\mathbf{A}_i$	Eq. (14)	$[N, 60]$
5	correlations $\mathbf{C}_i^{[q]}$	Eqs. (15)–(16)	$[N, 60]$
6	centre state $\mathbf{h}_i^{(0)}$	Eq. (17)	$[N, 240]$

Here L is the number of directed edges and N the number of atoms.

2 Stage 0 — Irrep bookkeeping

```
for l in range(l_max + 1):
    output_mul = hidden_dim if l == 0 else hidden_dim // (2 if l == 1 else 4)
    corr_mul = correlation_channels if l == 0 else correlation_channels // (2 if l == 1 else 4)
    output_irreps.append((max(1, output_mul), (1, (-1) ** l)))
    correlation_irreps.append((max(1, corr_mul), (1, (-1) ** l)))
```

The parity assignment $(\ell, (-1)^\ell)$ is the *natural-parity* choice: $\ell = 0$ even, $\ell = 1$ odd, $\ell = 2$ even. In VMK’s terminology these are the *polar* (true) irreducible tensors, VMK 5.5(2); pseudotensors are discarded. Channel multiplicities taper with ℓ (16/8/4), because high- ℓ channels carry less signal per channel.

This truncation is not an arbitrary economy. It coincides exactly with the Gaunt selection rule $C_{\ell_1 0 \ell_2 0}^{L 0} = 0$ unless $\ell_1 + \ell_2 + L$ is even (VMK 5.6(9)): natural parity keeps precisely the couplings that survive for spherical harmonics of a single direction.

Object	Irreps	Dim
irreps_node (species embedding)	64x0e	64
irreps_sh (harmonics)	1x0e+1x1o+1x2e	9
irreps_correlation ($\mathbf{A}$, $\mathbf{C}$)	16x0e+8x1o+4x2e	60
irreps_out (centre state)	64x0e+32x1o+16x2e	240

3 Stage 1 — Radial basis $B_n(d)$

SmoothACERadialBasis is the product of a compact envelope and a Bessel basis, manuscript Eqs. (7)–(9):

```
def forward(self, distances):
    return self.cutoff(distances).unsqueeze(-1) * self.basis(distances)
```

The envelope is the quintic smoothstep, written in factored form to avoid catastrophic cancellation in float32 just inside $x = 1$:

```
envelope = (1.0 - x).pow(3) * (1.0 + 3.0 * x + 6.0 * x.pow(2))
return torch.where(distances < self.r_max, envelope, torch.zeros_like(envelope))
```

Algebraically this is

$$f_c(d) = 1 - 10x^3 + 15x^4 - 6x^5 \equiv (1 - x)^3(1 + 3x + 6x^2), \quad x = d/r_c, \quad (1)$$

with $f_c(r_c) = f'_c(r_c) = f''_c(r_c) = 0$. This C^2 property is what keeps forces continuous as a neighbour enters or leaves the list.

The Bessel part handles the $d \rightarrow 0$ limit analytically:

```
bessel = torch.sin(scaled) / safe_dist
bessel = torch.where(distances.unsqueeze(-1) == 0, self.freq / self.r_max, bessel)
return self.norm * bessel # norm = sqrt(2/r_max)
```

giving

$$B_n(d) = f_c(d) \sqrt{\frac{2}{r_c}} \frac{\sin(n\pi d/r_c)}{d}, \quad \lim_{d \rightarrow 0} \frac{\sin(n\pi d/r_c)}{d} = \frac{n\pi}{r_c}. \quad (2)$$

The normalisation satisfies $\int_0^{r_c} B_n^2 d^2 dd = 1$. Output shape $[L, 12]$.

4 Stage 2 — Spherical harmonics $Y_{\ell m}(\hat{\mathbf{r}}_{ij})$

```
self.sh = o3.SphericalHarmonics(self.irreps_sh, normalize=True,
                                normalization="component")
```

`normalize=True` divides by $|\mathbf{r}|$, so the result is a pure function of direction; the radial dependence lives entirely in stage 1. The "component" convention means $\sum_m |Y_{\ell m}|^2 = 2\ell + 1$, which relates to VMK 5.10(1) by

$$Y_{\ell m}^{\text{e3nn}} = \sqrt{4\pi} Y_{\ell m}^{\text{VMK}}. \quad (3)$$

Output shape $[L, 9]$.

5 Stage 3 — The edge tensor $\mathbf{a}_{ij}$

This is the core of the descriptor (`_density`):

```
sender = edge_index[0]
receiver = edge_index[1]
radial = self.radial_basis(edge_len) # [L, 12]
harmonics = self.sh(edge_vec) # [L, 9]
weights = self.radial_net(radial) # [L, 1792] <- per-edge TP weights
edge_features = self.tp_density(node_attrs[sender], harmonics, weights) # [L, 60]
```

Three points deserve emphasis.

(a) **The edge carries the *neighbour's* species.** `node_attrs[sender]` is e_j , not e_i . The central species enters separately in stage 6.

(b) **radial_net produces tensor-product *weights*, not features.** It maps the 12 Bessel coefficients to all 1792 weights of `tp_density`, *per edge*. The radial dependence therefore enters as a learned, distance-dependent coupling $R_{cq}^{(\ell)}[\mathbf{B}(d_{ij})]$. FullyConnectedNet is bias-free, so $\mathbf{B} \rightarrow 0$ at the cutoff forces $\mathbf{a}_{ij} \rightarrow 0$.

(c) **Only three coupling paths exist**, because the first input is all scalars:

path	path_shape
0e x 0e -> 0e	(64, 1, 16)
0e x 1o -> 1o	(64, 1, 8)
0e x 2e -> 2e	(64, 1, 4)

Scalar $\otimes \ell \rightarrow \ell$ is the trivial coupling, so the whole operation collapses to manuscript Eq. (12),

$$a_{ij,cm}^{(\ell,(-1)^\ell)} = Y_{\ell m}(\hat{\mathbf{r}}_{ij}) \sum_q R_{cq}^{(\ell)}[\mathbf{B}(d_{ij})] e_{j,q}. \quad (4)$$

Verified numerically: for every edge and every $\ell > 0$ the (c, m) matrix of `edge_features` has **rank 1** (largest ratio of second to first singular value 3.4×10^{-16} over 608 edges), and the recovered per-block constant is exactly $1/\sqrt{2\ell+1}$ — the Clebsch–Gordan factor $C_{\ell_1 0 \ell_2 0}^{L 0}$ of VMK 5.6(14).

6 Stage 4 — The neighbour density $\mathbf{A}_i$

```
density = torch.zeros(node_attrs.shape[0], self.irreps_correlation_dim, ...)
density.index_add_(0, receiver, edge_features)
```

A single scatter-add sums every incoming edge onto its receiver, $[L, 60] \rightarrow [N, 60]$:

$$\mathbf{A}_i = \sum_{j \in \mathcal{N}(i)} \mathbf{a}_{ij}. \quad (5)$$

Permutation invariance is automatic: addition does not depend on neighbour order. This object is the ACE *A-basis*.

7 Stage 5 — The correlation ladder

```
correlation = density
output = center + self.order_mix[0](correlation)
if self.correlation_order >= 3:
    correlation = self.contractions[0](correlation, density)
    output = output + self.order_mix[1](correlation)
if self.correlation_order >= 4:
    correlation = self.contractions[1](correlation, density)
    output = output + self.order_mix[2](correlation)
```

This is the left-associated recursion of Eqs. (15)–(16), $C_i^{[1]} = \mathbf{A}_i$ and $C_i^{[q+1]} = \text{TP}_{q+1}(C_i^{[q]}, \mathbf{A}_i)$:

q	expression	nominal body order
1	$\mathbf{A}$	2
2	$\text{TP}(\mathbf{A}, \mathbf{A})$	3
3	$\text{TP}(\text{TP}(\mathbf{A}, \mathbf{A}), \mathbf{A})$	4

Each `contractions[k]` is a `FullyConnectedTensorProduct` on $60 \rightarrow 60$ with 8768 *shared* weights — model parameters, in contrast to `tp_density`, whose weights are generated per edge. Each `order_mix[q]` is an `o3.Linear` projecting one order into the 240-dim output space, and the orders are *summed*, so the model can weight body orders freely.

Two structural facts about this ladder:

- `contractions[0]` is $\text{TP}(\mathbf{A}, \mathbf{A})$ — the *same* tensor in both slots. By VMK 3.1(27), $\{\mathfrak{M}_{J_1} \otimes \mathfrak{M}_{J_2}\}_{JM} = (-1)^{J_1+J_2-J} \{\mathfrak{M}_{J_2} \otimes \mathfrak{M}_{J_1}\}_{JM}$, which renders 47.2% of its weights (4136 of 8768) structurally unidentifiable. Confirmed: a large purely antisymmetric perturbation changes the output by 2×10^{-14} .
- Because indices repeat in $\sum_j \mathbf{a}_{ij}$, $C^{[q]}$ is *not* a pure $(q+1)$ -body term. VMK 3.2(22) makes this exact: for a single neighbour the entire ladder collapses to one $Y_{\ell m}$, so expressive gain from higher $q_{\max}$ requires at least two distinct neighbours.

8 Stage 6 — The centre state $h_i^{(0)}$

```
center = torch.cat((
    self.center_proj(node_attrs), # [N, 64] scalars
    node_attrs.new_zeros((node_attrs.shape[0], non_scalar_dim)) # [N, 176] zeros
), dim=-1)
```

The central species enters *only* through the $\ell = 0$ block: a linear map of $\mathbf{e}_i$, zero-padded for $\ell > 0$. This is physically necessary — a species label is a scalar and cannot seed an $\ell > 0$ feature without a direction — and it is what makes two atoms with identical neighbour densities but different central species distinguishable. The result is manuscript Eq. (17),

$$\mathbf{h}_i^{(0)} = \text{Center}(\mathbf{e}_i) + \sum_{q=1}^{q_{\max}} \text{Linear}_q(C_i^{[q]}), \quad [N, 240]. \quad (6)$$

9 Full data flow

```
edge_len [L] --> radial_basis --> [L,12] --> radial_net --> [L,1792] (TP weights)
|
edge_vec [L,3] --> sph. harmonics --> [L,9] -----+
|                                                     +--> tp_density --> a_ij [L,60]
z [N] --> emb --> [N,64] --> node_attrs[sender] -----+ |
| index_add_v
| | A_i [N,60]
| |
| | +-- order_mix[0] <-- C^[1] = A -----+
| | +-- order_mix[1] <-- C^[2] = TP(A,A)
| | +-- order_mix[2] <-- C^[3] = TP(C^[2],A)
| |
v |
center_proj -----+ + <-----+
|
h^(0) [N,240]
```

`forward` additionally returns `edge_features` and `cutoff` unchanged. These become the frozen keys and values of the attention block: the edge tensors are computed once and never updated, which is precisely what makes TRACE a *fixed-environment* architecture.

10 Two design consequences

Parameter concentration. The final layer of `radial_net` is $32 \times 1792 = 57,344$ parameters, **43% of the model**. Because `emb[sender]` takes only n_{spec} distinct values, the model depends on that layer and the species embedding only through $M[s, h, c] = \sum_q W_2[h, q, c] E[s, q]$, which has just $n_{\text{spec}} \times 32 \times 28$ degrees of freedom — 95.3% redundant for the three-species CsPbI₃ system.

Selection rule. The natural-parity truncation is exactly the Gaunt rule $\ell_1 + \ell_2 + L$ even. Every odd combination ($1, 1 \rightarrow 1$; $1, 2 \rightarrow 2$; $2, 2 \rightarrow 1$; $2, 2 \rightarrow 3$) was verified to evaluate to identically zero in e3nn’s real basis.

Source: `flashace/physics.py`, class `ACEV2Descriptor`. Companion documents:
`docs/TRACE-EQUATION-LEDGER.pdf` (equation status), `docs/ANGULAR-MOMENTUM-REFERENCE.md` (VMK
derivations), `docs/PEER-REVIEW.md` (defect analysis).